

(087) 654-6559
Dublin, Ireland
manya.singh@ucdconnect.ie

Manya Singh

PhD Researcher

Portfolio:
github.com/manya09singh
linkedin.com/in/manyasinghdatascience/

Ph.D. researcher in robust machine learning with an emphasis on **trustworthy** and **responsible** AI. Experience working for projects funded by the Irish National Transport Authority and Irish Research Council.

SKILLS

Tools and Languages	Python, Git, C++, MATLAB, Java, SQL, LaTeX, R Markdown
Quantitative Research	Responsible AI, Trustworthy Machine Learning, Predictive Multiplicity, Robust Machine Learning, Rashomon Sets, Ambiguity, Geo-spatial Analysis, Unsupervised learning, Human-Centric AI, GPS Data-Driven Analysis,
Communication	English, Hindi, Kannada

TECHNICAL EXPERIENCE

Research Assistant / ISCycle Nov 2025 — Present
University of Limerick Limerick, Ireland

- Created a data cleaning pipeline that uses GPS data from two sources (Google maps and PowUnity) to create a unified data set for bicycle trip analysis.
- Created an unsupervised learning framework for trip purpose imputation from location, time and cycle features for bicycle trips.

Teaching Assistant / Computer Science Sep 2024 — Present
Beijing University of Technology Beijing, China

- Assisted in designing lab tutorials, demonstrations, grading exams on multiple modules such as Machine Learning, Network Analysis, Data Science in Practice, Data Science in Python etc.

Research Assistant / SECOREA Sep 2024 — Sep 2025
University College Dublin Dublin, Ireland

- Created reference load profiles for different types of academic buildings in UCD to support creating a local energy community.
- Worked with 8 years of electricity load data at a 15-minute interval across 6 different academic and residential buildings to explore the segmentation of electricity usage based on building types.
- Published findings in the [International Conference for Big Optimization 2025](#)

Research Assistant / Bike Library Nov 2023 — Aug 2024
University College Dublin Dublin, Ireland

- Collected, cleaned, and processed over 10,000 electric bike trips from the GPS data of the Dublin Bike Library, creating a map-matching function for over 1 million road segments.
- Researched appropriate model features to indicate the importance of e-bike type in route choice analysis using logistic regression and random forest models.
- Published findings in the [Journal of Transport Geography](#)

Risk Analyst / Cyber Risk Jan 2021 — Aug 2022
Deloitte USI Bangalore, India

- Led the automation unit for end-to-end on-boarding of applications through the pipeline to the Production stage from SIT/DEV from a single Jira ticket trigger.
- Designed scalable python and SQL scripts for automated XML generations with automated validation against high volume heterogeneous metadata for over 7000 files.
- Created standardized XML validation and modification scripts in an agile environment, used across all applications saving up to 60% of resources spent on manual testing.

EDUCATION

Doctor of Philosophy, Computer Science, *University College Dublin* Sep 2024 — ongoing
Master of Science in Data and Computational Science, *University College Dublin* Sep 2022 — Sep 2023
Bachelor of Technology in Electronics and Communication, *Manipal Institute of Technology* Jun 2017 — Aug 2021

ACTIVITIES

Rethinking Music Genres: *a data-driven categorization of music* Aug 2023
Odor Identification: *predictive modeling using molecular descriptors.* Jun 2021